

AI Tools Mastery

4-Week Hands-On Course Curriculum

Prompting • Creative & Multimodal Tools • Automation & Agents • Responsible Use • Capstone Project

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Course Overview

Duration	4 weeks (3 sessions/week, 2 hours each = 24 contact hours) + approx. 4 hrs/week self-paced practice
Format	Live workshop-style sessions: 30 min concept → 75 min guided lab → 15 min review
Audience	Professionals, students, and founders with no prior AI background who want to use AI tools productively at work
Prerequisites	Comfortable with a web browser, email, and basic documents/spreadsheets. No coding required.
Capstone	An end-to-end AI-powered workflow or product built and demoed in Week 4

Learning Outcomes

By the end of the course, learners will be able to:

1. Write effective prompts using structured techniques (role, context, examples, constraints, iteration).
2. Choose the right AI tool for a task across text, image, audio, video, research, and code.
3. Build a working automation that connects AI to real apps (email, docs, spreadsheets, chat).
4. Evaluate AI outputs for accuracy, bias, and hallucination, and apply responsible-use practices.
5. Design, build, document, and present a complete AI-powered project.

Tool Stack

Instructor note: Tool names and features change quickly. Verify current versions, pricing, and free-tier limits before each cohort and swap equivalents where needed.

Category	Tools used in labs
Chat assistants	Claude, ChatGPT, Gemini
Research	Perplexity, NotebookLM, Elicit / Consensus
Images	Midjourney, DALL·E / GPT image tools, Adobe Firefly, Ideogram
Audio / Video	ElevenLabs, Suno, Runway, HeyGen, Descript
Productivity	Notion AI, Gamma, Canva AI, Microsoft Copilot / Google Workspace AI
Automation	Zapier, Make, n8n
Coding / App building	Cursor, Claude Code, GitHub Copilot, Replit, Lovable / Bolt
Custom assistants	Claude Projects, Custom GPTs, Gemini Gems

Week 1 — Foundations & Prompt Engineering

Theme: *How AI models work, and how to talk to them well.*

Session 1.1 — How AI Tools Actually Work

- What a large language model is (and isn't); tokens, context windows, training cutoffs
- Why models hallucinate; where they're reliable vs. unreliable
- Tour of the landscape: chat assistants, multimodal models, agents, image/audio/video generators

Lab: Ask the same 5 questions to Claude, ChatGPT, and Gemini. Compare accuracy, tone, and reasoning. Document one hallucination you catch.

Session 1.2 — Prompt Engineering Fundamentals

- The prompt anatomy: role → context → task → format → constraints → examples
- Zero-shot vs. few-shot prompting; chain-of-thought ("think step by step")
- Iterative refinement: critique, revise, and "ask the model to ask you questions"

Lab: Rewrite 6 bad prompts into strong ones. Build a personal prompt library (10 reusable templates) in a Notion page or shared doc.

Session 1.3 — Working with Documents, Data & Files

- Uploading PDFs, spreadsheets, and images; summarization, extraction, Q&A over documents
- Structured output: getting tables, JSON, and checklists back
- Grounded research with Perplexity and NotebookLM (citations, source checking)

Lab: Upload a 20-page report → produce an executive summary, a stakeholder FAQ, and a fact-check table with sources.

Week 1 Assignment (due start of Week 2)

Prompt Post-Mortem. Pick a real task from your work or studies. Complete it with AI, then write a one-page report covering: what you prompted, what failed, how you fixed it, and time saved.

Week 2 — Creative & Multimodal Tools

Theme: *Generating and editing images, audio, video, and visual content.*

Session 2.1 — AI Image Generation

- Prompt structure for images: subject, style, lighting, composition, camera, negative prompts
- Consistency techniques: reference images, style seeds, character sheets
- Tools compared: Midjourney vs. DALL·E vs. Firefly vs. Ideogram (text rendering, licensing)

Lab: Create a 5-image brand visual set (logo concept, hero image, 3 social posts) with a consistent style.

Session 2.2 — AI Audio & Video

- Voice cloning and text-to-speech with ElevenLabs; music with Suno
- Video: text-to-video (Runway), avatar presenters (HeyGen), editing by transcript (Descript)
- Ethics checkpoint: consent, deepfakes, disclosure

Lab: Produce a 60-second explainer video: AI-written script → AI voiceover → AI visuals → assembled and captioned.

Session 2.3 — AI for Documents, Slides & Design

- Gamma / Canva AI for decks; Notion AI and Workspace AI for docs and meeting notes
- Turning a chat conversation into a polished deliverable

Lab: Convert last week's report summary into a 10-slide presentation and a one-page infographic.

Week 2 Assignment

Content Kit. For a product, cause, or personal brand, deliver: 1 landing-page copy draft, 3 images, 1 short video or audio clip, and 1 slide deck. Include a "tools & prompts used" appendix.

Week 3 — Automation, Agents & AI-Assisted Building

Theme: *Moving from chatting with AI to systems that run without you.*

Session 3.1 — Custom Assistants & Knowledge Bases

- Building a Claude Project / Custom GPT / Gemini Gem with instructions and uploaded knowledge
- System prompts vs. user prompts; designing guardrails and tone

Lab: Build a custom assistant for a specific role (e.g., customer-support bot, onboarding guide, study tutor) with at least 5 reference documents. Test it with 10 realistic questions.

Session 3.2 — No-Code AI Automation

- Triggers, actions, and AI steps in Zapier / Make / n8n
- Common patterns: email → summarize → Slack; form → classify → spreadsheet; RSS → draft post
- Error handling, cost control, and human-in-the-loop approvals

Lab: Build a working automation: incoming emails or form submissions are summarized, categorized, and logged to a spreadsheet with a Slack/Teams alert.

Session 3.3 — AI-Assisted Coding & App Building

No coding background needed.

- "Vibe coding" with Cursor, Claude Code, Replit, or Lovable: describing what you want and iterating
- Reading and debugging AI-generated code at a basic level; asking the AI to explain itself
- When to build vs. when to use an existing tool

Lab: Build and deploy a simple web app (e.g., a calculator, quiz, or form-based tool) using only natural-language instructions.

Week 3 Assignment

Automation Case Study. Document your automation with a flow diagram, screenshots, a cost estimate per 1,000 runs, and a list of failure modes with mitigations. Submit your final project proposal (see Final Course Project).

Week 4 — Responsible Use, Strategy & Final Project

Theme: *Using AI well, evaluating it critically, and shipping a real project.*

Session 4.1 — Evaluation, Ethics & Responsible Use

- Hallucination detection, bias testing, verification workflows
- Privacy and data handling: what not to paste into AI tools; enterprise vs. consumer tiers
- Copyright, attribution, and disclosure; company AI policies

Lab: Red-team a classmate's custom assistant from Week 3. Produce a findings report with 5 issues and fixes.

Session 4.2 — Project Build Studio

- Full working session with instructor and peer support
- Office hours by track: content/creative, automation, custom assistant, app
- Mid-session checkpoint: demo a working slice to a peer

Session 4.3 — Final Project Presentations & Course Wrap

- 7-minute demos + 3-minute Q&A per learner/team
- Peer scoring and instructor feedback
- Personal "AI Operating System" plan: which tools you'll use weekly, and for what
- Certificate of completion

Final Course Project

Goal

Design and build a complete, working AI-powered solution to a real problem you care about, then present it.

Project Tracks (choose one)

Track	Deliverable	Example
Content Engine	A repeatable multimodal content pipeline	Weekly newsletter generated from 3 RSS feeds with images and an audio version
Custom Assistant	A deployed knowledge-based assistant with guardrails	HR policy bot for a small company, tested against 25 questions
Workflow Automation	A live automation with AI decision steps	Lead-qualification flow: form → score → CRM → personalized email draft
AI-Built App	A deployed web app built with AI coding tools	Habit tracker, budget planner, or internal request portal
Open Track	Instructor-approved combination	Research-to-report pipeline for a grad student

Required Components

1. **Problem statement** — who it's for, what pain it solves, how success is measured
2. **Working build** — live demo, not slides
3. **Architecture diagram** — tools used and how data flows
4. **Prompt & configuration appendix** — every prompt, system instruction, and automation step
5. **Evaluation** — at least 10 test cases with pass/fail, plus one documented hallucination or failure and how you handled it
6. **Responsible-use notes** — data privacy, disclosure, and limitations
7. **Reflection** — time saved, cost, what you'd build next

Timeline

Milestone	When
Proposal submitted (one page)	Week 3, Session 3.3
Proposal feedback returned	Week 4, Session 4.1
Build studio + checkpoint demo	Week 4, Session 4.2
Final presentation and submission	Week 4, Session 4.3

Grading Rubric (100 points)

Criterion	Points
Solves a real, clearly defined problem	15
Works as demonstrated (functionality & reliability)	25
Appropriate tool selection and prompt quality	15
Evaluation rigor (test cases, failure analysis)	15
Responsible-use considerations	10
Documentation and reproducibility	10
Presentation clarity	10

Assessment Summary

Component	Weight
Weekly assignments (3)	30%
Lab participation & prompt library	15%
Peer red-team report	10%
Final project	45%

Pass mark: 70%. Certificate awarded on pass + final project demo.

Materials & Setup Checklist

Send to learners before Week 1.

- Accounts created: one chat assistant (paid tier recommended), Perplexity, one image tool, Zapier or Make, one AI coding tool
- Shared workspace: Notion or Google Drive folder for the prompt library and assignments
- Sample datasets and documents provided by instructor (report PDF, CSV of leads, sample emails)
- Budget guidance: approx. \$40–60 in tool subscriptions for the month; free tiers work for most labs
- Community channel (Slack/Discord) for questions between sessions

Optional Extension Modules

For 5- or 6-week variants of the course:

- AI for data analysis (spreadsheet AI, Python via AI notebooks)
- Building with APIs and basic agent frameworks
- AI in sales, marketing, HR, or education (role-specific deep dives)